IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 11 December 2025, accepted 3 February 2026, date of publication 20 February 2026, date of current version 5 March 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3666781

SURVEY

Human or Machine? A Survey on Machine-Generated Text Detection

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This work was supported in part by the Instituto Politécnico Nacional under Grant 20251107; and in part by the the Secretaría de Educación, Ciencia, TecnologÍa e Innovación de la Ciudad de México, through the Project "Aplicación del Cómputo Urbano Para Analizar la Dinámica Urbana y la Sustentabilidad de las Grandes Ciudades," under Grant CM-SECTEI/197/2025.

ABSTRACT As generative AI advances rapidly across education, research, medicine, and journalism, machine-generated text (MGT) raises questions about authenticity, ethics, and social impact. To ground this discussion, we conducted a linguistic analysis, covering phonology, morphology, syntax, semantics, lexicon, and pragmatics, which uncovers robust MGT signatures, such as lower perplexity and simpler morphology. We then consolidate the state-of-the-art by reviewing 30 benchmark corpora, totaling over 4 million text samples, and 44 empirical studies, including outcomes from six major shared tasks. Detection approaches are grouped into five broad classes: classical machine learning, deep learning, transformer-based architectures, commercial AI detectors, and statistical tools. Transformer models achieve near-perfect accuracy (≈100%), while human evaluators peak at ≈77% accuracy. This survey also highlights common evaluation setups and the core performance measures used to assess model effectiveness. Future MGT detectors can become truly fair, scalable, and effective across languages and domains by expanding corpus diversity and innovating resource-efficient, adversarially robust methods.

INDEX TERMS Generative AI, large language models (LLMs), machine-generated text detection, multilingual corpora.

I. INTRODUCTION

Large Language Models (LLMs) like GPT-4 [1] and LLaMA [2] are primarily aimed at text generation tasks and have been shown to generate contextually relevant and semantically accurate paraphrases [3]. LLMs power applications such as chatbots and virtual assistants [4], enabling tasks like content creation and report generation [5]. LLMs have greatly enhanced the automation of generating articles, reports, and other types of text, resulting in significant time and resource savings. They can be used across multiple domains, improving content generation in a wide range of fields. In customer service, LLMs like ChatGPT have improved response speed and quality, often providing more efficient and consistent replies than human agents [6]. In healthcare, LLMs assist in generating clinical documentation, medical reports, and educational content [7], [8]. In journalism, these models support content creation and newsroom automation by generating relevant articles and analyzing large amounts of data [9]. LLMs are also applied in software development for code generation and completion. A study by Nejjar et al. [10] evaluated several LLMs and identified GPT-3.5 [11], GPT-4, and YouChat[1] as the best performers, excelling at code generation.

Despite the numerous advantages of LLMs, their use also presents several challenges. In academia, concerns about academic dishonesty and plagiarism have emerged. In healthcare, the potential risks associated with generating inaccurate or misleading content highlight the need for careful oversight. Similarly, in journalism, the use of LLMs has raised concerns

The associate editor coordinating the review of this manuscript and approving it for publication was Asadullah Shaikh.

[1]https://web.youchat.com Last visited: 03-18-2025

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about the role of new journalists and ethical issues, such as the spread of fake news. In code generation tasks, another concern is the potential for errors and inaccuracies. Tambon et at. [12] identified issues such as syntax errors, incorrect code generation, and hallucinations in code generated by leading LLMs like OpenAI Codex.[2] Furthermore, fake machine-generated reviews also raise trust issues among people regarding decision-making while buying products or visiting restaurants and tourist destinations [13], [14].

LLMs are becoming increasingly advanced in text generation, blurring the line between human-written and machine-generated text (MGT). A new task, MGT detection, has emerged to address this challenge, aiming to identify whether a given text is human-written or machine-generated. As a result, it is essential to develop a robust and efficient detection system to maintain the authenticity of human-created content by preventing plagiarism and avoiding the spread of fake news and misinformation. The detection of MGT has been of great concern, leading to many workshops, conferences, and shared tasks focused on differentiating human-written and LLM-generated text, such as AuTexTification [15],[3] ALTA [16],[4] CLIN33 [17],[5] MLMAC [18],[6] SemEval-Task8 [19],[7] DAGPap24 [20],[8] and GenAI Content Detection [21].[9] These tasks not only focus on distinguishing human-written from MGT but also include sub-tasks like model attribution, boundary detection, and adversarial and nonuniversal cross-domain identification of MGT. Similarly, PAN at CLEF 2025 [22][10] has announced the shared task "Generative AI Detection"[11] focusing on the diverse uses of MGT. The task is divided into two sub-tasks: 1) classifying the text as written by a human or machine, and 2) categorizing the text into one of six classes: (i) Fully human-written, (ii) Human-initiated, machine-continued, (iii) Human-written, machine-polished, (iv) Machine-generated, then human-refined (obfuscated), (v) Machine-generated, then human-edited, (vi) Deeply mixed text. The classes dive deep to cater to all possibilities in the suspicious text, such as text being initiated by humans and further polished by machines, or vice versa.

The development of reliable detection systems heavily depends on the availability of high-quality Corpora. A well-constructed Corpus is vital for the model's ability to generalize and identify the nuances of machine-generated content in diverse contexts and languages, including English, German, French, Spanish, and many others.

[2]https://openai.com/index/openai-codex/ Last visited: 03-18-2025
[3]https://sites.google.com/view/autextification Last visited: 02-06-2025
[4]http://alta.asn.au/events/sharedtask2023 Last visited: 01-20-2025
[5]https://sites.google.com/view/shared-task-clin33 Last visited: 02-06-2025
[6]https://mlmac.io/ Last visited: 02-06-2025
[7]https://github.com/mbzuai-nlp/SemEval2024-task8 Last visited: 02-06-2025
[8]https://sdproc.org/2024/sharedtasks Last visited: 02-06-2025
[9]https://genai-content-detection.gitlab.io/ Last visited: 01-18-2025
[10]https://pan.webis.de/clef25/pan25-web/ Last visited: 02-06-2025
[11]https://pan.webis.de/clef25/pan25-web/generated-content-analysis Last visited: 02-06-2025

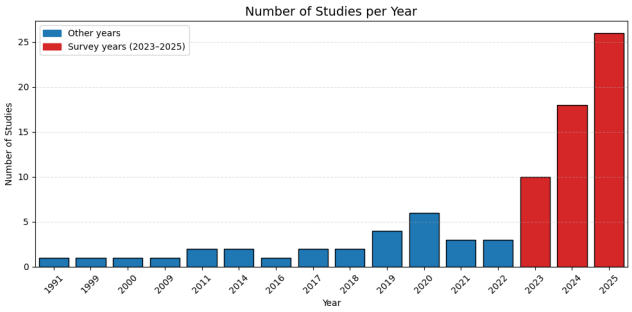


FIGURE 1. Year-wise distribution of studies included in the survey.

Furthermore, given the wide range of LLMs available, such as GPT-3, GPT-4, and LLaMA, the detection system must be capable of recognizing the various text patterns and stylistic features generated by these models. These models often exhibit unique linguistic characteristics, such as fluency and coherence, as well as subtle inconsistencies that differ from human writing styles. MGT is often modified using techniques like paraphrasing, text summarization, translation, content expansion, and even the injection of errors [23], [24]. These modifications make the detection task more challenging, as they hide the original structure and style of the machine-generated text, making it harder to distinguish from human-written content. As LLMs continue to advance, the boundaries between human-written and machine-generated text become increasingly blurred, further complicating the task of detection. In some cases, human-written text may be edited or enhanced by LLMs, resulting in hybrid content that poses additional challenges. Moreover, the multilingual capabilities of LLMs introduce additional complexity, as models trained on various languages can generate text in multiple linguistic forms and structures. To address these challenges, detection systems must adapt and continuously improve, incorporating new techniques that can recognize advanced manipulations and evolving LLM-generated text patterns across different languages.

We aim to provide a comprehensive review of recent advances in MGT detection, focusing on studies published between 2023 and 2025. Figure 1 presents the number of studies published each year, with the selection of studies limited to those within this timeframe to highlight the current state-of-the-art in the domain. This survey addresses the importance of MGT detection and the challenges involved, such as the difficulty in distinguishing human-written text (HWT) from MGT due to subtle differences in syntax, semantics, and style. We categorize existing Corpora according to language, domain, and resource availability, offering insights into their structure and utility for training detection models. Additionally, we examine the range of detection methods, from traditional machine learning approaches to transformer-based models and commercial AI detectors. The survey also assesses the effectiveness of these methods using standard evaluation metrics such as accuracy, F1-score, precision, and recall. Moreover, we highlight the

limitations of current studies and discuss potential areas for improvement, aiming to guide future research in MGT detection.

The contribution of this survey is manifold. Given the extensive work already done in the domain, this survey consolidates the state-of-the-art research and identifies potential areas for improvement. First, this survey focuses on an in-depth analysis of the linguistic differences between HWT and MGT, drawing from various studies. Second, it explores the development of corpora for MGT detection across different languages, analyzing trends in Corpus creation and usage. Finally, the survey also presents the most common learning settings, highlighting evaluation methodologies and performance metrics, both automatic and human-based, used to assess the effectiveness of MGT detection models. Through this comprehensive approach, we aim to highlight key insights and gaps in the current research landscape.

This survey is systematically organized as follows: Section II highlights the objectives of the survey. Section III introduces a unifying linguistic framework, distinguishing formal and pragmatic features that differentiate HWT from MGT. Section IV reviews the corpora that power research in this area. This section is further divided by exploring the corpora for English-language IV-A, low-resource languages IV-B, and multilingual corpora IV-C. Section V discusses the methodologies used in text generation and detection, from classical machine learning to state-of-the-art transformer-based models. This section also explores the top-ranked studies in different shared tasks V-A. Section VI discusses the most commonly used learning settings and evaluation setups. Section VII details the human and automated evaluation measures used for the task. The practical implications and real-world use cases of MGT detection are presented in Section VIII, followed by a broader discussion in Section IX. Section X outlines the limitations of current approaches. Finally, Section XI discusses the conclusion of this survey.

II. RESEARCH OBJECTIVE

This section states the aims of our survey: to map where machine-generated text (MGT) detection stands today and what is needed next. We review detection methods, the corpora that support them, and the evaluation practices used to assess performance. By connecting current knowledge with open challenges, we aim to give researchers and practitioners practical guidance for building reliable detectors that keep pace with rapid model advances. The four objectives are:

1) Review linguistic signals that separate human-written text (HWT) from MGT, covering syntactic, lexical, and semantic differences reported across studies.
2) Classify existing corpora by language setting (English, multilingual, low-resource) and summarize size, domain, language variety, and the generators used to produce machine text.
3) Compare detection methods spanning classical machine learning, deep learning, transformer-based models, commercial AI detectors, and statistical approaches, noting strengths, limits, and typical failure modes.
4) Describe evaluation practice including common data splits (train–test, k-fold), metrics (accuracy, F1, precision, recall, AUROC), and good practices for fair, robust assessment (e.g., leakage control, cross-domain tests, robustness checks).

Concurrently with our work, several 2025 surveys cover related ground. Cao et al. [25] review AI-generated content across unimodal and multimodal settings, track historical trends, and list open problems, but do not focus on detection methods or multilingual corpora. Kwon et al. [26] broaden the scope to "fake text" by combining misinformation and LM-generated text, emphasizing societal impact rather than corpus-level comparisons or deployment-oriented evaluation. Wu et al. [27] is closest to ours: a methods-centric review of watermarking, statistical and neural detectors, datasets, and challenges such as distribution shift, adversarial attacks, and weak evaluation practice. In contrast, our survey (i) provides a corpus-by-language synthesis with explicit inclusion/exclusion criteria, (ii) proposes a unified taxonomy linking generators, transformations, detectors, and evaluation, and (iii) places special emphasis on multilingual and low-resource gaps. These additions complement prior work and offer practice-ready guidance for research and deployment.

By addressing these objectives, we give a concise picture of the current state of MGT detection and point to concrete opportunities for future work.

III. THEORETICAL FRAMEWORK: A TAXONOMY OF LINGUISTIC FEATURES

Understanding the difference between human-written text (HWT) and machine-generated text (MGT) is critical for many reasons, such as proving content authenticity, catching plagiarism, and fighting misinformation. Even as we work on creating better models for detection [27], [28], it's just as important to understand what makes these two types of text distinct. This section lays out a clear framework to help us systematically analyze those differences, pulling from what we know about computational linguistics and authorship attribution. We've broken the linguistic characteristics down into two key categories: Formal Features and Pragmatic Features. This structure gives us the unifying framework the reviewers asked for and helps us better discuss how these features affect detection performance. Table 1 presents the overview of linguistics feartures.

"Human-written text refers to content written by humans, e.g., students, teachers, or researchers, reflecting their ideas and knowledge."

vs

"Machine-generated text describes content produced by LLMs, e.g., GPT-4, LLaMA, etc., that mimic human writing."

A. FORMAL LINGUISTIC FEATURES

Formal features are the quantifiable, low-level aspects of language that you can measure with a computer. They provide the core data for many detection models.

1) PHONOLOGY, MORPHOLOGY, AND WORD CHOICE

MGT often has phonological patterns and simpler word structures that are different from HWT. A few studies have shown that MGT uses certain sounds like approximants and nasals more often than human text [29]. When you look at word structure, HWT tends to have more variety, using things like prepositions and verbs more frequently. MGT, on the other hand, relies more on coordinating conjunctions, nouns, and pronouns [29], which can make sentences feel grammatically correct but a bit rigid. When it comes to vocabulary, MGT tends to use more difficult, content-heavy words, probably because it's trained on massive datasets that favor being informative over being easy to read.

2) SYNTAX AND PERPLEXITY

Syntax also shows a clear difference. MGT often produces shorter sentences with a wider range of syntactic structures [30]. HWT, however, is more likely to use prepositional phrases to create longer and more complex sentences [30]. This suggests that while LLMs are great at creating grammatically correct variety, they don't always mimic the complex thinking that goes into human-written sentences. This leads to a key difference, i.e., perplexity. Perplexity measures how unpredictable a text is. MGT generally has a lower perplexity score because it's designed to be coherent and predictable. Human writing naturally has more variation and is harder to predict. We've seen that while newer, more advanced models like GPT-3 have lower perplexity than earlier ones, some models like TinyLlama are getting better at being unpredictable, which makes their text harder to detect [31].

B. PRAGMATIC AND NARRATIVE FEATURES

Pragmatic features are the more subtle, high-level parts of language that show things like intent, emotion, and context. These are the aspects that machines still struggle to get right.

1) EXPRESSIVENESS AND EMOTIONALITY

Machines really struggle to show genuine emotion or a unique voice. While you can prompt an LLM to sound emotional, the result often feels neutral and impersonal, lacking the organic, context-dependent emotionality of human writing [30]. Human text naturally includes a sense of personal perspective and emotional nuance that MGT just can't replicate, making it feel more alive and authentic.

2) NARRATIVE AND COHESION

Human writing is great at building a story or a coherent argument across an entire document. We use things like context-dependent references and link ideas together in a way that feels natural. MGT, on the other hand, often focuses on delivering information and can struggle with keeping a consistent, long-range narrative or making clear connections between ideas [32]. This is especially true in things like creative writing or persuasive essays, where that deeper human-level cohesion is essential.

C. UNIFYING PERSPECTIVE

When you look at all these features together, they form a complex picture. As LLMs get better, they're starting to mimic formal features like vocabulary and syntax more effectively. But they still really struggle with the more subtle, subjective pragmatic features, such as true emotional depth and cohesive narrative flow. This is a critical takeaway for MGT detection; going forward, models will need to get better at capturing these nuanced pragmatic elements if they want to stay ahead.

IV. CORPORA FOR MACHINE-GENERATED TEXT DETECTION

Corpora play a crucial role in Natural Language Processing (NLP) research by providing structured data for training and evaluating models. In the domain of MGT detection, benchmark Corpora are vital to ensure that detection models perform well across different languages and real-world contexts. While a variety of Corpora exist for detecting AI-generated text in English, there is a noticeable gap in resources focusing on low-resource languages for the MGT task, such as Arabic and Chinese. Despite these languages being well-resourced for other NLP tasks [33], [34], they remain low-resource for the MGT detection task. The advent of multilingual corpora helps bridge this gap, enabling detection models to handle multiple languages effectively. These corpora expand the scope of MGT detection beyond English, providing valuable resources for developing more robust models that can identify machine-generated content in diverse linguistic settings. This section explores the process of corpus creation and highlights the corpora available for MGT detection in English, low-resource languages, and multilingual contexts.

In this survey, we include corpora released in 2023–2025 that explicitly address MGT detection and either introduce a new dataset or extend an existing one with human vs. machine labels. Eligible papers must describe the generator/model and the prompt or editing setup, and provide language and domain information. We exclude resources without transparent labeling or a clearly defined detection task. Candidate corpora were identified through targeted venue searches and citation chaining, with only those meeting these criteria retained.

A. ENGLISH CORPORA

The NLP research community has predominantly focused on English for MGT detection, primarily due to the widespread availability of large-scale English Corpora. English is widely used in AI model training and testing, which has accelerated

TABLE 1. Overview of linguistic features.

Feature	Definition	Categories/Levels	Example
Phonology	Study of sounds that deal with the formation of words, which influence pronunciation	Phonemes, Allophones, Stress, Intonation	The phonemes "c" and "b" used in "cat" and "bat" differentiate their pronunciation.
Morphology	Study of the structure and formation of words	Morphemes (roots, prefixes, suffixes), Inflection, Derivation	The word "unhappiness" consists of three morphemes: "un-" (prefix), "happy" (root), and "-ness" (suffix).
Syntax	Deals with the combination of words and the structure of sentences	Part of Speech (POS), phrase and sentence structure	"She loves cake." In this sentence, "she" is the subject, "loves" is a verb, and "cake" is the object.
Semantics	Study of meaning in language (in context)	Sentiment Analysis, Named Entity Recognition (NER)	"What do you do?" This sentence can mean asking a question or describing an action.
Lexicology	Study of words related to dictionaries	Etymology, Word Length, Word Frequency	Compute how many times "water" appears in an essay.

the development of detection models. Several English Corpora have been introduced for detecting MGT, including those from diverse domains such as academic writing, news articles, and social media [35], [36]. These Corpora have contributed significantly to the progress of MGT detection systems. Although there are significant similarities among them, such as the common use of LLMs like GPT for generating text and the contrast between HWT and MGT, each Corpus presents unique methodologies and focuses.

The **CHEAT** corpus [37] primarily targets scientific articles, specifically abstracts from IEEE Xplore.[12] It consists of 15,395 human-written abstracts and 35,304 machine-generated abstracts produced using ChatGPT (GPT-3.5-turbo). The machine-generated texts are categorized into three classes: generated, polished, and mixed. The ''generated'' abstracts are entirely produced by ChatGPT based on titles and keywords, while the ''polished'' abstracts involve revisions of human-written texts. In the ''mixed'' category, polished abstract sentences are randomly replaced with human-written abstract sentences that were semantically the same. This approach to categorizing MGT based on how it was generated (entirely vs. modified) is a distinctive feature of the CHEAT corpus.

The **Maktabdar** Corpus [38] is developed to support MGT detection, focusing on academic question–answers from the fields of computer science, artificial intelligence, and cybersecurity. Human-written answers were sourced from technical encyclopedias, while MGT responses were created via ChatGPT using the same prompts. The corpora were proposed for the article and sentence levels. The article-level corpus comprises 1,018 article-level samples, further consisting of balanced ChatGPT-generated and human-authored answers. The sentence-level corpus, obtained by extracting sentences from the articles, comprises 4,008 MGT and 3,336 HWT, totaling 7,344 entries. Rigorous cleaning was applied to ensure consistency across both categories.

The **ICNALE** corpus [39] is designed to explore the impact of author-level attributes on MGT detection. It consists of 5,138 human-written essays from the ICNALE corpus, paired with 61,656 LLM-generated texts created by 12 models like Qwen2.5, LLaMA3, and Mistral. Each sample is annotated with demographic metadata including gender, CEFR proficiency, academic discipline, and language environment (EFL, ESL, native) that no prior data contains. This Corpus, totaling 66,794 texts, provides a rich basis for evaluating detector performance across diverse author profiles and testing for potential detection biases.

The **SFR** [40] is a new short-form review corpus for MGT detection. They scraped genuine reviews from two Indian e-commerce sites (Flipkart, Nykaa), then generated synthetic reviews with two large language models (Google Gemini and Mistral). They extracted 67,641 raw reviews; after filtering to at least 25 words, 26,581 remained. For the generation of MGT, 17,645 reviews were written to be casual (20–80 words), with product specs, price, rating, and sentiment randomized to mimic real posts. The final set had 43,226 reviews across tech, cosmetics, home, and other categories. The data are not publicly released due to privacy.

The **FakeGPT** corpus [35] is developed for fake news detection, but with a focus on comparing real and generated fake news. This corpus contains 5,200 instances, balanced between human-written and ChatGPT-generated fake news selected from nine publicly available corpora for fake news, including FAKENEWSNET [41], LIAR [42], and a synthetic news corpus using ChatGPT. For the generation of news through ChatGPT, methods such as altering text and inventing stories are used, as ChatGPT does not generate harmful information due to its moderation mechanism. The quality of the generated news was then analyzed by ChatGPT itself and by human evaluation. Notably, ChatGPT achieved 72% accuracy in identifying fake news, whereas human evaluators identified fake news with an accuracy of just 33%. This study not only assesses the effectiveness of detection models but also explores how LLMs compare to human evaluators in spotting fake content.

The **Mixed** corpus [43] adopts a different approach by creating a mixed collection of HWT and MGT from various domains, including email, news, game reviews, academic paper abstracts, speeches, and blogs. The corpus is divided into 3,600 instances, with 2,400 being MGT and the remaining 1,200 being HWT. It features three types of text manipulation, including polishing, rewriting, and completion

[12]https://ieeexplore.ieee.org Last visited: 03-18-2025

on HWT. The polishing operation is used to alter the word and sentence. The rewrite operation extracts the key information from the original text and then rewrites the text using that information. In completed text instances, one-third of the original text is taken, and the LLMs are prompted to complete the text using the given text. Both rewriting and completion operations are performed at the paragraph level. MGTs are humanized by introducing typos or adapting them to reflect typical human writing habits. This corpus is notable for the varied techniques applied to both HWT and MGT, creating a rich Corpus for analyzing the nuances of machine-generated writing. To perform all these tasks, two LLMs, GPT-4 and LLaMA2 [2], are used for each task.

The **Myle Ott** corpus [31] is a review-based Corpus that uses the Myle Ott opinion spam corpus as a source [44], consisting of a subset of 800 reviews, half of which are positive and the other half negative, aimed at detecting fake reviews. Positive reviews are sourced from TripAdvisor, while negative reviews come from platforms such as Expedia, Hotels.com, and Yelp. These reviews are then used to generate fake reviews. Machine-generated reviews are produced using several models, including GPT-3 (DaVinci), GPT-2, TinyLlama, and a T5-based paraphrase model, with each model generating 800 reviews. The final corpus consists of 4,000 reviews,800 human-written and 3,200 machine-generated.

The **AiGen-FoodReview** corpus [36] is created to detect fake restaurant reviews and includes data from 9,200 restaurants in New York City from SafeGraph.[13] This multi-modal Corpus analyzes not only fake reviews but also images. It contains 10,143 human-written reviews paired with at least one image collected from elite Yelp users, ensuring high quality. For machine-generated reviews, both text and images were created. Text reviews were generated using GPT-4-Turbo, while images were produced with DALL-E 2 [45]. A total of 10,001 synthetic review–image pairs were generated, balancing the corpus between authentic and generated pairs. The final corpus contains 20,144 review–image pairs, making it one of the most comprehensive multi-modal corpora for fake review detection. The generated reviews had higher linguistic complexity, while the images showed greater brightness, saturation, and clarity compared to the authentic data.

[46] create a **GPT-2 XL** corpus by sourcing data from four different domains to ensure diversity. The corpus consists of news articles, Q&A, prompts, and movie reviews extracted from the XSum [47], SQuAD [48], WRITING [49], and IMDB [50]. A sample of 5,000 texts is selected from each corpus and labeled as HWT. An additional sample of 5,000 texts is chosen to generate MGT using the GPT-2 XL model, with inputs of 30, 60, and 90 tokens to produce sentences with a length of 512. The final balanced corpus consists of 10,000 texts,5,000 labeled as HWT and 5,000 labeled as MGT.

[13]https://www.safegraph.com/ Last visited: 03-20-2025

The **MAGE** corpus [51] provides a more general-purpose resource for MGT detection. It draws from a variety of domains, including news, reviews (SciXGen [52]), QA (SQuAD [48]), story writing, Wikipedia, and research article abstracts. This corpus contains 154,078 human-written texts and 294,381 machine-generated texts, created using 27 different LLMs, including GPT, LLaMA, and others. The texts are generated using a variety of prompting techniques, including completing partial texts or generating content based on specific topics. The breadth and diversity of the MAGE corpus make it one of the largest and most varied Corpora for MGT detection, offering rich data across multiple domains.

The **GPABENCHMARK** corpus [53] was built by gathering English academic abstracts from three key disciplines, i.e., computer science, physics, and humanities/social sciences. Human-written abstracts were sourced from top-tier conference proceedings, arXiv postings, and Springer SSRN, while the machine-generated texts are produced by ChatGPT (GPT-3.5-turbo) using a range of prompt types. The MGT was generated using three prompting techniques, like prompting the model to write complete text or asking it to complete or polish the given text. On the basis of these generation approaches, the corpus is organized as fully written (GPT-WRI), completed (GPT-CPL), and polished (GPT-POL) abstracts. In total, GPABench2 comprises roughly 2.38 million samples, with approximately 2.23 million GPT-generated and 150,000 human-written abstracts.

The **ALTA** corpus [16] includes four generative benchmark corpora from multiple domains: machine translation (WMT [54]), summarization (CNN-DailyMail [55]), language pre-training (Wiki-Data), and OpenWebText [56]. A sample of 10,000 HWT was gathered directly from these sources to generate MGT. LLMs such as GPT-2, GPT-3.5-turbo, and GPT-4 were used to generate 10,000 instances of MGT. The final corpus contains 20,000 balanced instances, with 10,000 labeled as HWT and 10,000 labeled as MGT.

The **Yelp-based** corpus [57] adopted a different approach by focusing on customer reviews from Yelp. In total, 177,410 reviews are collected from 5,959 restaurants, with 46,144 being elite and 131,266 being non-elite reviews. The OpenAI GPT-3 (Curie and Davinci) is used to generate machine-generated reviews, with the model temperature adjusted to influence the randomness of the text generation derived from the sample elite reviews. The final corpus is obtained by balancing authentic and fake reviews, forming a total of 24,000 reviews. This Corpus, while similar to the FakeGPT corpus in its focus on fake reviews, specifically targets the restaurant review domain and includes a substantial number of elite Yelp reviews as the basis for generating fake content.

B. CORPORA FOR LOW-RESOURCED LANGUAGES

Low-resource languages, such as Urdu, pose unique challenges in MGT detection due to the limited availability of annotated Corpora. While some languages, such as Arabic and Chinese, have extensive resources for general NLP

tasks [33], [34], they remain low-resourced in the MGT detection task. Linguistic differences, such as syntactic complexity and morphological variation, further complicate detection. The lack of large-scale, high-quality Corpora restricts model development and evaluation. This section presents notable corpora developed for low-resource languages.

Shamardinaet al. [58] proposed **CoAT**, a balanced Russian corpus for MGT detection with 123,241 human-written (HWT) and 123,241 machine-generated (MGT) texts across six domains (Russian National Corpus slices, news, Wikipedia, personal diaries, government/strategic documents, and social media). Machine texts cover multiple generation types, including machine translation with back-translation, paraphrasing, simplification, summarization, and open-ended generation, produced by 13 models (OPUS-MT, mBART-50, M2M-100; ruGPT-2 Large; ruGPT-3 Small/Medium/Large; ruT5-base-multitask/-Large; mT5 Small/Large; ruT5-base-sum-Gazeta; MBART-ru-sum-Gazeta). The authors de-duplicate, filter for Russian, remove profanity, and set task-specific length ranges to avoid length cues.

Ali et al. [24] presented MGT detection corpora for the Urdu language named **HLU**. Human-written instances are extracted from Wikipedia on 13 categories, including Sports, Countries, History, and Natural Disasters. For each category, 26 articles were collected, for a total of 338 HWT instances. MGT instances were generated using the API of GPT-3.5-turbo and GPT-4-turbo by using the concept of in-context learning [59]. Moreover, the models were self-criticized for generating repetitive words, making errors in facts, and writing incoherent text. For each model, 338 MGT instances were created, making the total size of the corpus 1,014 instances, including the HWT and MGT. The Corpus was generated for different granularity levels like document, paragraph, and sentence. The paragraph- and sentence-level instances were created using the document-level instance by extracting sentences and paragraphs.

Alhayan et al. [60] constructed an Arabic **fake review detection corpus** by translating English Amazon reviews [61] into Arabic using the Google Translate API.[14] Two LLMs, ULMFiT [62] and GPT-2, are used to generate machine-generated reviews. The corpus consists of various product categories, including Pet Supplies, Home and Kitchen, Electronics, Sports, Tools and Home Improvement, Clothing, Shoes, and Jewelry, and Movies, resulting in a Corpus of 40,440 reviews,20,224 human-written and 20,261 machine-generated.

Alshammari et al. [63] developed two Arabic corpora: **"Large Corpus"** and **"Custom Corpus"**, addressing linguistic complexities such as diacritics. The **"Large Corpus"** contains 43,958 instances, comprising 21,979 HWT and 21,979 MGT. HWT are gathered from three existing sources, Arabic-SQuAD [64], TyDiQA [65], and MLQA [66]. MGT is generated using English samples from the HC3 [67] corpus, translated into Arabic. The **"Custom Corpus"** incorporates 1,050 HWT from the Tashkila corpus [68], 249 from the Shamela[15] Library, and 249 from the Aldiwan[16] website. ChatGPT 3.5, ChatGPT 4.0, and Bard [69] each generate 513 MGT instances, leading to a final Corpus of 1,539 balanced HWT and MGT samples.

Wang et al. [70] introduced a **Chinese corpus** for detecting AI-generated text at both document and sentence levels, addressing the challenges of in-domain overfitting and out-of-domain (OOD) generalization by providing two test sets. The in-domain test set is of the same domain as the train set, i.e., Q/A, while the OOD test set is taken from the news domain. At the document level, 12,853 questions from the Chinese sample of the HC3 corpus [67] are used to generate responses from nine different LLMs, including GPT, which are labeled as MGT, while the original human-written responses from the HC3 corpus are used as HWT. At the sentence level, 5,589 instances are sampled. The Corpus consists of 151,799 samples, including 21,681 human-written and 96,453 AI-generated texts in the training set. The in-domain test set contains 3,000 human-written and 26,750 MGT instances, while the OOD test set includes 2,000 HWT and 1,915 MGT instances.

Irrgang et al. [71] introduced a **German corpus** for AI-generated text detection, sourcing texts from various domains, including journalistic texts (die Tageszeitung), everyday language from blog discussion threads (Zeit Online), encyclopedic texts (Wikipedia), scientific texts (Springer Nature),[17] and literary texts (Wikimedia).[18] To ensure authenticity, all texts were collected before 2016, before the emergence of LLMs. MGT is generated using LLaMA 2, GPT-3.5, Snoozy [72], Wizard [73] and Mistral [74].[19] A total of 5,000 text samples are collected from each domain for HWT, and an equal number is generated by LLaMA. GPT-3 produces 800 instances per genre, while Mistral generates only 471. Snoozy generates 4,162 samples for Zeit Online, with fewer samples produced for other genres. As a result, the corpus is imbalanced, containing 25,000 HWT and 36,681 MGT instances.

Nitu et al. [75] introduced a **Romanian corpus** for AI-generated text detection in digital libraries. The Corpus comprises 59,096 documents, including 7,387 HWT and 51,709 MGT, covering five domains: books, news, medical, legal, and scientific articles. MGT is generated using RoGPT2 [76], GPT-Neo-Ro,[20] DaVinci-003 [77], Flan-T5 [78], Opus-MT [79] and mBART [80]. Various text generation methods,such as text completion, back-translation, and paraphrasing,were employed to

[14]https://translate.googleapis.com Last visited: 01-18-2025

[15]https://shamela.ws/ Last visited: 01-18-2025

[16]https://www.aldiwan.net/ Last visited: 01-18-2025

[17]https://www.springernature.com/de Last visited: 01-18-2025

[18]https://dumps.wikimedia.org/other/static_html_dumps/current/de/ Last visited: 01-18-2025

[19]https://huggingface.co/malteos/hermeo-7b Last visited: 03-20-2025

[20]https://huggingface.co/dumitrescustefan/gpt-neo-romanian-780m Last visited: 03-20-2025

produce machine-written content. The Corpus maintains balance within each domain, with human-written and machine-generated samples collected equally. The document counts are as follows: 1,401 for books, 4,320 for news, 557 for medical, 1,000 for legal, and 109 for scientific articles.

C. MULTILINGUAL CORPORA

Multilingual corpora extend the capabilities of MGT detection by allowing models to generalize across various languages. These corpora support cross-lingual studies, enabling the evaluation of detection performance in different linguistic contexts. Unlike monolingual Corpora, multilingual corpora address the challenges of linguistic diversity, but also introduce issues such as data imbalance and varying text generation patterns across languages. While some corpora focus on high-resource languages, others aim to provide balanced Corpora across diverse language families. In this section, we explore key multilingual corpora used in MGT detection.

Fivez et al. [17] developed a corpus for the shared task **CLIN33** in English and Dutch. The human-referenced text is collected from five domains, including news articles, tweets, product reviews, poetry, and columns ''open source'' domain for English. MGT is generated by ChatGPT, GPT-4, and Vicuna-13B [81].[21] The total human and LLM-generated instances for the English language were 715 each. Similarly, a balanced ratio of human and machine-generated instances was generated for Dutch as well; each contains 750 instances.

Gryka et al. [82] introduced a **Gryka Email** corpus for detecting AI-generated emails in English and Polish, consisting of 21,468 samples, with 10,356 human-written and 11,112 AI-generated emails. English data includes 9,885 human-written and 10,618 AI-generated emails, while Polish data consists of 471 human-written and 494 AI-generated emails. The AI-generated samples are created by paraphrasing original human-written emails using OpenAI's GPT-3.5-turbo, preserving the tone and content of the original emails. The corpus includes linguistic features like perplexity, sentence variability, and typographical errors, essential for building detection models.

Dugan et al. [23] proposed the **RAID** corpus, one of the largest corpora to date, with 6 million instances. The HWT instances are extracted from 11 different corpora across domains such as news, Wikipedia, recipes, poems, reviews, etc. MGT instances are generated using 11 different models, including MPT [83], Mistral, LLaMA, Cohere [84], ChatGPT, GPT-2, GPT-3, and GPT-4, with additional adversarial attacks like paraphrasing and altering sentence structures to increase difficulty. The final corpus consists of 14,971 HWT and 509,014 MGT instances.

Wang et al. [85] proposed the **M4** corpus for the SemEval shared task, aiming to detect machine-generated texts across multiple languages, domains, and generation methods in a black-box setting. The corpus includes diverse multilingual data from several languages. MGT is generated using six LLMs: Davinci003, ChatGPT, GPT-4, Cohere, Dolly-v2 [86], and BLOOMZ [87]. The LLMs are prompted to generate articles, abstracts, and answers corresponding to HWT, which are collected from 11 different sources on various topics. Given its combination of multiple generators, domains, and languages, the corpus is categorized as multi-generator, multi-domain, and multilingual. This structure enables robust evaluations across cross-lingual, cross-domain, and cross-generator settings, addressing the need for resources that facilitate generalized detection across diverse contexts. The corpus contains 147,000 samples, with 35,798 HWT and 97,753 MGT, generated as follows: Davinci003 (23k), ChatGPT (32k), GPT-4 (14k), Cohere (13k), Dolly-v2 (14k), and BLOOMz (14k).

Schaaff et al. [88] developed a multilingual **Human-AI-Generated Text (HAIGT)** corpus to classify human-generated, AI-generated, and AI-rephrased texts in English, French, German, and Spanish. The corpus consists of 100 samples per language for each type of text, across ten topics: biology, chemistry, geography, history, IT, music, politics, religion, sports, and visual arts, offering balanced representation across languages and topics. By balancing language and topic coverage, this corpus provides a robust benchmark for evaluating models designed to detect and distinguish between different types of generated content.

Macko et al. [89] developed the **MULTITuDE** corpus for MGT detection across multiple languages. HWT originates from the MassiveSumm [90][22] Corpus, and MGT is generated using models like Alpaca-LoRA [91], GPT-3.5-turbo [92], GPT-4, LLaMA, OPT [93], and VICUNA. The final corpus consists of 7,992 human-written and 66,089 machine-generated instances. While detectors are trained in English, Spanish, and Russian, they were tested across 11 languages. The proposed corpus focuses exclusively on the news domain, with careful attention given to ensure that no harmful or misleading information is included in the data.

Guo et al. [67] introduced the **Human ChatGPT Comparison Corpus (HC3)**, a Corpus designed to compare ChatGPT-generated responses with those of human experts across various fields, including medicine, finance, law, and psychology. The corpus includes 40,000 questions and 125,230 responses, consisting of 80,805 human-generated and 44,425 ChatGPT-generated responses. The Corpus is available in English and Chinese. The English portion contains 24,322 questions, 58,546 human responses, and 26,903 responses from ChatGPT, while the Chinese portion includes 12,853 questions, 22,259 human responses, and 17,522 responses from ChatGPT. To create the Corpus, ChatGPT is prompted with the same questions answered by human experts, with additional tailored prompts to align the AI's responses with human styles. The corpus is organized to facilitate direct comparisons between human and AI responses, making it ideal for side-by-side evaluations. Its

[21] https://lmsys.org/blog/2023-03-30-vicuna Last visited: 02-08-2025

[22] https://github.com/danielvarab/massive-summ Last visited: 02-20-2025

multilingual and multi-domain nature supports the development of effective AI text detectors and provides a valuable resource for assessing the fluency, neutrality, and accuracy of AI-generated content.

Corizzo et al. [94] introduced a bilingual **Corizzo Essay** corpus designed to evaluate AI-generated essay detection using one-class learning models like One-Class Support Vector Machines (OCSVMs) [95] and Local Outlier Factor (LOF) [96]. The corpus consists of 1,370 essays, including 685 human-written and 685 AI-generated ones, in English and Spanish. Human essays are sourced from the Uppsala Student English Corpus [97][23] and the UC Davis Corpus of Written Spanish.[24] AI-generated essays are created by prompting ChatGPT with the same essay topics. This corpus focuses on linguistic features such as repetitiveness, readability scores, emotional semantics, and part-of-speech tagging, enabling detection without relying on labeled AI-generated data for training. The English portion includes 335 human-written and 335 AI-generated essays, while the Spanish portion contains 350 human-written and 350 AI-generated essays. The corpus addresses the challenges in detecting AI-generated essays, particularly in multilingual academic contexts, making it a valuable resource for studying authenticity and academic integrity in L2 writing.

Sarvazyan et al. [15] proposed a corpus for the **AuTexTification** shared task in English and Spanish, covering five domains: Legal, News, Reviews, Tweets, and How-to articles. The corpus includes 27,688 human-written texts, and machine-generated texts are generated using various models, including BLOOM and GPT. The final corpus contains 56,010 instances, making it a versatile resource for evaluating detection systems across multiple languages and domains.

Table 2 summarizes the available corpora, detailing their key attributes relevant to MGT detection. For each Corpus, the table includes information on the number of human-written and machine-generated instances, the models used to generate MGT, the languages covered, and the domains from which the data was sourced. This summary offers valuable insights into the scope and diversity of multilingual Corpora for MGT detection research.

V. METHODS

As LLMs continue to evolve and generate text that is increasingly difficult to distinguish from human writing, it becomes crucial to develop efficient and reliable methods for detecting MGT. These methods play a key role in preserving the authenticity of online content and addressing concerns such as plagiarism, misinformation, and the ethical use of AI technologies.

The challenge lies in identifying the subtle differences between HWT and MGT, which often requires innovative techniques to analyze patterns in language, syntax, and meaning.

In this section, we explore the approaches developed to address this growing issue. Detection systems range from classical machine learning models to state-of-the-art transformer models, all designed to achieve the highest possible accuracy. In addition, statistical tools (e.g., Fast-DetectGPT and GLTR) and linguistic insights are being combined to create robust and adaptable systems that can handle the ever-evolving capabilities of LLMs across different contexts and applications.

Wang et al. [98] introduce BENATTEN, an attention-centric detector built on the hypothesis that MGT pays attention to tokens in a more natural way than human text, yielding stronger conformity to Benford's law. BENATTEN converts text to attention maps, compresses them, tests first-digit Benford patterns, merges with attention stats, and uses a small classifier to label AI vs. human. Against strong criterion-based and model-based baselines, for example, GLTR, DetectGPT, OpenAI detector, BENATTEN outperformed on the Cgtd, CHEAT, and HC3 corpora by obtaining 99% accuracy.

Torres et al. [99] proposed a multi-task and multilingual detector with a shared RemBERT encoder and two outputs: first decide whether a text is human-written or machine-generated, and the other identify the language, either English or Spanish. Training both tasks together helps the shared representation, improving transfer across languages and robustness to new domains, without adding extra time at prediction. The model is fine-tuned on the AuTexTification data and evaluated by language and domain, plus two additional Spanish test sets. It beats the shared-task baseline: for Spanish, the ''human'' class F1 score rises from about 54% to about 74.5%, and the overall F1 score averaged across classes improves by roughly nine percentage points; English gains about one percentage point. On the external Spanish sets (Coco Dolly ES and Coco Twitter ES), performance stays strong, showing good generalization to writing styles not seen during training.

Shamardina et al. [58] compared a term frequency–inverse document frequency (TF–IDF) linear classifier with fine-tuned transformer models (ruBERT, ruRoBERTa, XLM-RoBERTa, and RemBERT) for Russian machine-generated text detection. They trained and evaluated all models on balanced splits, reported accuracy and macro-F1 scores, and analyzed results by generation type and by domain. The best overall model was ruRoBERTa, which achieved approximately 87% accuracy and macro-F1. By generation type, open-ended generation and summarization were nearly perfect (up to about 99%), paraphrase and simplification were moderate (about 77%), and machine translation was the most difficult (about 61% macro-F1). Human evaluators achieved about 66% accuracy.

Jadhwani et al. [40] used simple, interpretable features like TF–IDF, cosine similarity between the product title

[23]http://ota.bodleian.ox.ac.uk/repository/xmlui/handle/20.500.12024/2457 Last visited: 03-01-2025

[24]https://github.com/ucdaviscl/cowsl2h Last visited: 02-28-2025

TABLE 2. Comparison of Machine-Generated Text Corpora. H/M refers to the number of human-written and machine-generated samples.

Corpus	H/M	LLM	Language(s)	Domain
CHEAT [38]	15k/35k	2	English	Computer science research articles
Maktabdar [39]	3k/4k	2	English	QA from computer science, artificial intelligence, and cybersecurity
ICNALE [40]	5k/61k	2	English	Essay
SFR [41]	26.6k/17.6k	2	English	E-commerce product reviews
FakeGPT [36]	2.6k/2.6k	1	English	News
MIXSET [44]	1.2k/2.4k	2	English	Email, news, game reviews, abstracts, TED Talk, blogs
Myle Ott [32]	800/3.2k	4	English	Hotel reviews
AiGen-FoodReview [37]	10k/10k	2	English	Restaurant reviews
GPT-2 XL [47]	5k/5k	1	English	News articles, Q&A pairs, prompts, and movie reviews
ALTA [16]	10k/10k	3	English	General
Yelp-based [58]	12k/12k	2	English	Restaurant reviews
MAGE [52]	154k/294k	27	English	Reviews, news, QA, story writing, Wikipedia, abstract research articles
GPABench2 [54]	150k/2235k	27	English	Research article abstracts
CoAT [59]	123k/123k	13	Russian	RNC, News, Wikipedia, Diaries, Gov/Strategic, Social Media
HLU [24]	338/676	2	Urdu	Sports, Countries, History, Natural Disasters, etc.
Fake review detection [61]	20k/20k	2	Arabic	Reviews
Large Corpus [64]	21k/21k	1	Arabic	Q&A
Custom Corpus [64]	1539/1539	1	Arabic	Q&A
Chinese [71]	26k/125k	9	Chinese	Web Q&A, encyclopedias, etc.
German [72]	25k/36k	4	German	Fake news, Wikipedia, scientific publications, etc.
Romanian [76]	7k/51k	6	Romanian	Books, scientific articles, medical, etc.
HAIGT [89]	100/200	1	English, French, German, and Spanish	Biology, chemistry, geography, etc.
Gryka Email [83]	10k/11k	1	English and Polish	Emails
M4 [86]	35k/97k	6	English, Chinese, Russian, Arabic, Urdu, Indonesian, and Bulgarian	Wikipedia, Reddit ELI5, etc.
RAID [23]	14k/509k	11	English, Czech, and German	News, abstracts, recipes, Reddit, book summaries, poetry, reviews, Wikipedia
MULTITuDE [90]	7k/66k	8	English, Spanish, Russian, Portuguese, Catalan, German, Dutch, Ukrainian, Czech, Arabic, and Chinese	News
HC3 [68]	80k/44k	1	English and Chinese	Q&A
Corizzo Essay [95]	685/685	1	English and Spanish	Essays
AuTexTification [15]	27k/56k	3	English and Spanish	Legal, News, Reviews, Tweets, and How-to articles
CLIN33 [17]	1.4k/1.4k	3	English and Dutch	News articles, tweets, product reviews, poetry, etc.

and the review, VADER sentiment, word and character length, and counts of parts of speech and emojis. They then trained classical machine learning models including logistic regression, random forest, support vector machine, and XGBoost using a balanced corpus of 30,000 instances. The random forest performed best, achieving 95% accuracy using all features. A lighter XGBoost with only the statistical features reached 90.21% with much lower compute. The most useful signals were cosine similarity, sentiment, and length; parts-of-speech and emoji counts added little.

Yu et al. [37] trained transformer-based models (DistilBERT, BERT, RoBERTa, BERT-multilingual, and PubMedBERT) on the CHEAT Corpus for MGT detection. They also evaluated existing text synthesis detectors, including Grover [100],[25] ZeroGPT,[26] OpenAI-detector [101],[27] ChatGPT-detector-roberta,[28] and ChatGPT-qa-detector-roberta.[29] Models were trained on generated, polished, and mixed abstract subsets and evaluated using accuracy and AUC. Transformer-based models achieved near-perfect AUC, with all models reaching 100% on the generated subset, while PubMedBERT outperformed others on polished (99%) and

[25]https://originality.ai/blog/grover-ai-content-detection-review Last visited: 03-20-2025

[26]https://www.zerogpt.com/es Last visited: 03-20-2025

[27]https://huggingface.co/spaces/openai/openai-detector Last visited: 03-20-2025

[28]https://www.aimodels.fyi/models/huggingFace/chatgpt-detector-roberta-hello-simpleai Last visited: 03-20-2025

[29]https://www.eligeia.com/modelos/huggingface/hello-simpleai/chatgpt-qa-detector-roberta Last visited: 03-20-2025

mixed (87%) subsets. In contrast, AI detectors performed poorly, except ChatGPT-qa-detector-roberta, achieving only 97% on the generated subset.

Ali et al. [24] trained and fine-tuned the XLM-RoBERTa model on the HLU Urdu corpus, achieving an F1-score of 84.5% at the paragraph level and 70.6% at the sentence level. The model performed better at the paragraph level, likely because the larger text context provided more distinguishing features for accurate classification. In contrast, human evaluation yielded F1-scores of 81.7% and 82.6% at paragraph and sentence levels, respectively, highlighting the challenge of differentiating human- and LLM-generated text. The study highlights the need for more corpora in underrepresented languages, such as Urdu, to improve text detection models.

Oghaz et al. [38] employed classical machine learning, deep learning, and transformer models to classify ChatGPT-generated content using the proposed academic corpus for both article and sentence levels. Classic classifiers, including Naive Bayes, SVM, Random Forest, KNN; deep models like BiLSTM; and transformer models including DistilBERT, RoBERTa, and a custom RoBERTa variant, were evaluated under various hyperparameters. The best results were achieved on the proposed custom variant of RoBERTa, obtaining near-perfect accuracy of 99%. The study also compared these models with Turnitin's AI detector, which achieved 95% accuracy on article-level data.

Li et al. [39] conducted an out-of-domain evaluation of AI-generated text detectors using the proposed corpus. The study applied both classifier-based detectors like RoBERTa-Base (GPT-2), RoBERTa-Large (GPT-2), RoBERTa-Base (ChatGPT), RADAR, and metric-based detectors including GLTR, Binoculars, Fast DetectGPT, and DetectGPT, analyzing performance using accuracy at fixed false positive rates. The Binoculars detector achieved over 94% accuracy on low-proficiency (A2/B1) English learners, showing the highest performance but also revealing strong bias based on language proficiency and environment.

Koike et al. [102] utilized the M4 Corpus in this study to support detection of LLM-generated content. It contains 3,000 pairs of human-written and AI-generated texts across four domains (Wikipedia, Reddit, WikiHow, arXiv) and three LLMs (ChatGPT, GPT-4, Dolly-v2). The Corpus is divided into training, validation, and test sets in a 2,000/500/500 split for each domain–generator pair. This study introduced ExaGPT equipped with BERT-large, an interpretable LLM text detector that mimics human decision-making by comparing each text span with nearest neighbors from a datastore of known human-authored and machine-generated spans. The method segments input text into meaningful spans using a dynamic programming algorithm, scoring them based on similarity, length, and label distribution. The proposed ExaGPT was then compared with RoBERTa, LR-GLTR, and DNA-GPT. ExaGPT achieved 61% on human evaluation and 90% to 99% on other different subsets of data.

Zhou et al. [103] utilized a subset of the M4 Corpus to evaluate the DP-Net model for AI-generated text detection. The Corpus includes over 30,000 samples across five domains (Wikipedia, Reddit, WikiHow, PeerRead, and arXiv) and texts generated by three LLMs: ChatGPT, Davinci003, and Flan-T5. Human-written and LLM-generated texts are paired across each domain, with sample sizes per domain reaching up to 6,000. The proposed DP-Net is designed to achieve both high generalization and robustness. The method uses dynamic perturbations added to text embeddings, generated through reinforcement learning (DDPG), to simulate domain shifts. A noise-enhanced embedding is processed alongside the original to enforce similarity and stability during training. The model is optimized using a composite loss function balancing classification accuracy and embedding consistency. DP-Net is trained on arXiv ChatGPT data and tested across diverse domains, achieving superior accuracy under standard and adversarial conditions such as synonym substitution and paraphrasing attacks. To evaluate the effectiveness of DP-Net compared with a Naive Classifier, RoBERTa, SCRN, GLTR, and Fast-DetectGPT, DP-Net outperformed by achieving accuracy of 86% for the generalization task and, to check robustness, obtained 91% and 68% for synonym substitution and paraphrase attacks.

Min et al. [104] used HC3, which includes document-level and sentence-level samples in both English and Chinese, generated by ChatGPT. Additional out-of-domain Corpora include OpenGPT-WebText (GPT-3.5-turbo), POGER (GPTs, LLaMA-2, Vicuna, Alpaca), ChatGPT-CNews (GPT-3.5-turbo in Chinese), and Polished Abstract (ChatGPT-polished scientific abstracts in Chinese). This multilingual corpus spans diverse generators and domains, enabling the model to generalize across languages, domains, and LLM variants. The proposed bilingual detection framework combines semantic features from XLM-RoBERTa's CLS vector with statistical metrics like log-rank, token probability, and cumulative probability for fine-grained feature fusion. The model integrates these features through a normalization module before classification and employs SHAP to interpret feature contributions. Experiments show strong performance of the proposed method across both in-domain and out-of-domain settings, compared to baselines like GLTR and DetectGPT, achieving 99% and 95% accuracy for in- and out-of-domain settings.

Huang et al. [35] used ChatGPT to detect fake news using the FakeGPT corpus. The study identified nine reasons for fake news and applied ChatGPT with and without reason-aware prompts, incorporating these reasons. While ChatGPT showed promising performance on some corpora, there remains room for improvement.

Vijini et al. [31] first identified the most discriminative words for the generated category (''unhelpful'', ''terrible'', ''delicious'', ''outdated'', ...), while words for the non-generated category were more related to objects, places, or personal pronouns, like ''door'', ''floor'', ''coffee'',

''Michigan'', and ''she''. This showed that the vocabulary and style of generated reviews differ significantly from authentic ones, making fake reviews easier to detect. The authors then fine-tuned transformer-based models, including BERT, SciBERT, XLNet, and ELECTRA, on the Myle Ott corpus. Among these models, BERT was found to be the most effective in detecting generated content.

Gambetti et al. [36] directly tested OpenAI RoBERTa [101] and fine-tuned GPTNeo and BERT on the text reviews from the AiGen-FoodReview corpus. They trained CvT-13, ViT-B/16 and ResNet-50 on the training set of images from the same corpus. The study also introduced a multi-modal approach using CLIP [105] and FLAVA [106]. CLIP uses a fusion-free approach, while FLAVA employs the late fusion method. Additionally, logistic regression and random forest models were trained using handcrafted features. For the text reviews, readability, complexity, and perplexity features were selected, while only one image feature,photographic attributes,was chosen.

Lee et al. [46] fine-tuned and analyzed the BERT-base model on the corpus generated by different models separately. They utilized standard fine-tuning and proposed adversarial fine-tuning of the LLM. Moreover, different split ratios were used for training and testing, such as 8:2, 5:5, and 2:8, to critically analyze the model's performance in different scenarios. The best results were achieved with the proposed adversarial fine-tuning for the WRITING corpus,the split ratio of 8:2 and an accuracy of 95.03%.

Alhayan et al. [60] proposed an ensemble learning framework on the proposed corpora. The authors experimented with several machine learning classifiers, including Logistic Regression, Random Forests, Decision Trees; deep learning models such as CNN, RNN, and LSTM; and transformers such as AraBERT and DistilBERT. A train–test–validation split of 80%, 10%, and 10% was used to ensure a robust evaluation process. Metrics such as accuracy, precision, and recall were applied to measure classifier performance. Among the machine learning models, Logistic Regression delivered the best accuracy of 86.85%. In the deep learning models, the best accuracy achieved in ensemble settings was 89.57%. Overall, AraBERT outperformed all the models, obtaining an accuracy of 90%, akin to that of the combination of LR and CNN, which also provides 89.70%.

Alshammari et al. [63] fine-tuned transformers, i.e., AraELECTRA and XLM-R, on the proposed large and custom corpora. The corpora were split into train–test–validation to ensure a comprehensive evaluation. AraELECTRA and XLM-R were fine-tuned on both corpora, and each is evaluated on its corresponding validation and test sets. In addition, the fine-tuned models were also evaluated in cross-settings, i.e., models fine-tuned on the large corpus were tested on the custom test set, and vice versa. The proposed models were evaluated on AIRABIC[30] and outperformed the existing AI detectors (GPTZero[31] and OpenAI Text Classifier[32]), achieving 81% accuracy. Additionally, incorporating a dediacritization layer before the classification model significantly improved the detection accuracy. This step notably boosted the classification accuracy, increasing it from 81% to as high as 99%, and in some cases even reaching 100%.

Wang et al. [70] tackled the challenge of detecting AI-generated Chinese text, addressing the complexities of domain shifts and diverse linguistic characteristics. This study employs a statistical-based classifier (Fast-DetectGPT, GLTR, PPL), zero-shot classifiers (ChatGPT, GPT-4), and supervised classifiers (BERT, RoBERTa, LLaMA, Mistral). The evaluation was conducted on both in-domain and out-of-domain test sets, where the best results were achieved by instruction-tuned LLMs: 98.52% accuracy on in-domain data and 96.70% on out-of-domain data. The findings emphasized the versatility of instruction-tuned LLMs, demonstrating their ability to generalize across corpora with varying linguistic properties.

Irrgang et al. [71] investigated the impact of fine-tuning transformer models using generator-specific corpora enriched with features derived from comprehensive linguistic and stylistic analyses. The study refined several models, including DistilBERT, RoBERTa, BERT, LLaMA, and Mistral, by optimizing them with evaluation metrics such as F1-score, AUC, Precision, and Recall. Among these, DistilBERT emerged with slightly superior performance, achieving an F1-score of 0.94, while both RoBERTa and BERT recorded an F1-score of 0.93.

Nitu et al. [75] implemented two primary detection methods, a transformer-based model and a classic machine learning model. The transformer model, which leverages the Romanian pre-trained RoBERTa-base, utilizes token-based text representations for multi-class classification. It employs techniques such as gradient accumulation and early stopping to prevent overfitting. The classic machine learning approach combines linguistic complexity indices, such as robustness and lexical diversity, with an XGBoost classifier. Both models achieved competitive performance, with the transformer-based method excelling in the legal and news domains by achieving F1-scores of 90% and 96%, respectively. Similarly, the machine learning–based method outperformed in books, medical, and scientific domains and obtained F1-scores of 90%, 96%, and 91%.

Gryka et al. [82] analyzed linguistic features in AI-generated emails, focusing on text imperfections and stylistic anomalies. Machine learning models like Decision Tree, Random Forest, MLP Classifier, K-nearest Neighbors, Support Vector Classifier, AdaBoost, Gaussian Naïve Bayes, and Quadratic Discriminant Analysis were trained on these extracted features, achieving the best F1-score of 98.88% for English emails and 92.11% for Polish emails using Random

[30]https://github.com/Hamed1Hamed/AIRABIC Last visited: 01-18-2025

[31]https://gptzero.me/ Last visited: 01-18-2025

[32]https://beta.openai.com/ai-text-classifier Last visited: 01-18-2025

Forest. The study concluded that text imperfections, such as repetitive patterns and unnatural phrasing, significantly distinguish AI-generated emails from human-authored ones.

Zhang et al. [43] developed metrics- and model-based detectors using the proposed mixed corpus. The metrics-based models include entropy, GLTR, DetectGPT, log-rank, and log-likelihood. In contrast, model-based approaches include RADAR, DistilBERT, OpenAI Classifier, GPTZero, Ghostbuster, GPT-Sentinel, and ChatGPT Detector, which were also evaluated. Experimentation on mixed text obtained the best F1-score of 87% on the RADAR model, while the pure HWT and MGT were also classified, and GPT-Sentinel, RADAR, and DistilBERT achieved 100% accuracy.

Dugan et al. [23] employed 12 detectors, including different neural-based, statistical, and commercial models. Neural includes RADAR and three different variants of RoBERTa. Metrics-based models include GLTR, Binoculars, Fast DetectGPT, and LLMDet, and commercial detectors such as GPTZero, Originality, Winston, and ZeroGPT. The neural models,RoBERTa fine-tuned on the GPT-2–generated data, RADAR, and commercial GPTZero,performed well across all domains. The best result achieved by Binoculars was around 99% accuracy. The detectors were evaluated using accuracy at a false positive rate (FPR) of 5%. The study revealed that open-source detectors alarmingly marked HWT as MGT. To mitigate this effect, the repetition penalty reduced the accuracy. Different attacks were introduced in the corpus, so, surprisingly, some detectors performed well on one set but poorly on the other.

Li et al. [51] used the MAGE corpus to assess a range of detection techniques for machine-generated texts. This study set up eight distinct testbeds by combining different configurations of fixed, arbitrary, and unseen domains and models, and the eighth testbed exclusively evaluated the impact of paraphrasing attacks. The study compared several methods, including FastText, GLTR, DetectGPT, and a Longformer-based detector, and found that Longformer consistently achieved superior performance, with average recall rates ranging from 68% to 96% across all test scenarios.

Liu et al. [53] extensively explored the online, pre-trained, statistical-based detectors and fine-tuned language models. The study also proposed a deep learning framework named CheckGPT. The approach used a two-stage design: a pre-trained language model (RoBERTa-large) was employed to encode text into rich representations, and a lightweight classification head, based on LSTM with attention, aggregated these features for final prediction. The online detectors include ZeroGPT, GPTZero, and OpenAI's detector, which showed modest or poor performance, especially for machine-generated abstracts. Seven statistical models, including Rank and Log-Rank, were also investigated. The proposed CheckGPT outperforms both traditional hand-crafted linguistic models that used NELA features with Gradient Boosting and full-model fine-tuning of BERT, DistilBERT, RoBERTa, and GPT-2. In experiments, it consistently achieves detection accuracies above 99% for fully written abstracts, with slightly lower but still exceptional performance for completed and polished abstracts. The framework also demonstrates excellent transferability across prompts, disciplines, and academic writing sections, as well as robust resistance against various adversarial modifications.

Gambetti et al. [57] examined fake machine-generated restaurant reviews (Yelp-based corpus) by employing different classical machine learning, deep learning, and LLMs. This study also included human evaluation and statistical analysis. The CML models are Logistic Regression (LR), Naive Bayes (NB), Random Forest (RF), and XGBoost (XGB), which achieved the maximum accuracy of 85.07% on LR. The BiLSTM outperformed LR and achieved 93.71% accuracy. LLMs like GPT-2 and GPTNeo were also fine-tuned on the proposed corpus. GPTNeo obtained the highest accuracy of 95.51%.

Schaaff et al. [88] addressed the scalability of machine learning techniques for multilingual classification tasks, focusing on English, French, German, and Spanish corpora. The researchers used classical machine learning classifiers like Random Forest, XGBoost, and Multi-Layer Perceptron (MLP). In addition, features such as perplexity, semantic features, lookup lists, readability, etc., were added to enhance model interpretability. Evaluation metrics such as accuracy and F1-score were applied. The best F1-scores achieved by Random Forest for English, French, German, and Spanish texts are 98%, 95%, 97%, and 99%, respectively, for the detection of the MGT task. The study demonstrated that traditional machine learning models remain competitive for multilingual tasks, especially when supported by robust feature engineering.

Gou et al. [67] investigated the stylistic and linguistic differences between ChatGPT-generated texts and those written by human experts using the proposed HC3 corpus. Logistic transformer-based models like RoBERTa and GLTR were used for classification. RoBERTa achieved the highest F1-score, up to 96%.

Corizzo et al. [94] employed one-class methods to detect repetitive patterns and stylistic features unique to AI-generated essays. The approach focused on minimal supervision, leveraging extracted linguistic features for classification. The best performance on English was achieved with an F1-score of 95% on an autoencoder for repetitive content settings, underscoring the potential of one-class models to handle novel and underrepresented corpora in detection tasks, while for Spanish, the linear kernel obtained an F1-score of 72% for readability settings.

Reference [89] developed and evaluated several black-box (GPTZero[33] and ZeroGPT), statistical (log-likelihood [101], log-rank, DetectGPT [107], and rank, GLTR [108]), and fine-tuned (mGPT, mBERT, MDeBERTa, GPT-2, ELECTRA and RoBERTa) detectors. Among the black-box detectors, ZeroGPT achieved the highest accuracy at 47%. The statistical models performed well with accuracies between

[33]https://app.gptzero.me/ Last visited: 01-18-2025

TABLE 3. Summary of Machine-Generated Text Detection Methods. CML, DL, Trans, EL, CAID, and SA refer to classical machine learning, deep learning, transformer-based models, ensemble learning, commercial AI detectors, and statistical approaches, respectively.

Reference	CML	DL	Trans	EL	CAID	SA	Best Results
[99]			✓		✓	✓	BENATTEN achieved SOTA (AUC 99.9 on Cgtd; F1 99.7 on HC3) and is robust to paraphrase/noise.
[100]			✓				RemBERT achieved an F1-score of 74.5% on the Spanish "human" class.
[59]	✓		✓				ruRoBERTa achieved 87% accuracy.
[41]	✓					✓	Random Forest performed best, achieving 95% accuracy.
[38]			✓		✓		Transformer models achieved 100% accuracy on generated subset; PubMedBERT performed best on others.
[24]			✓				XLM-RoBERTa achieved F1-score of 84%.
[39]	✓	✓	✓		✓		Custom RoBERTa obtained 99% accuracy.
[40]			✓			✓	Binoculars achieved over 94% accuracy.
[103]			✓			✓	ExaGPT achieved 61% human eval. and 90–99% on other subsets.
[104]	✓	✓	✓			✓	Proposed model: 86% accuracy, 91/68% under synonym/paraphrase attacks.
[105]		✓				✓	Model obtained 99% and 95% accuracy (in/out-domain).
[36]			✓				ChatGPT with reason-aware prompt achieved 84%.
[32]			✓				BERT-base scored 99% F1 on subset generated with LLaMA.
[37]	✓		✓				Multimodal FLAVA achieved 99% accuracy.
[47]			✓				BERT-base achieved 95% accuracy.
[61]	✓	✓	✓	✓			AraBERT obtained 90% accuracy.
[64]			✓		✓		AraELECTRA and XLM-R achieved up to 100%.
[71]			✓		✓	✓	Qwen achieved 98.5% (in-domain), 96.7% (out-of-domain).
[72]			✓				DistilBERT achieved F1 = 0.94.
[76]	✓		✓				Transformer models: F1 90–96%; CML best on other domains.
[83]	✓						Random Forest: F1 = 98% (EN), 92% (PL).
[44]			✓		✓	✓	GPT-Sentinel, RADAR, and DistilBERT achieved 100%.
[23]			✓		✓	✓	Binoculars achieved 99% accuracy.
[58]	✓	✓	✓			✓	GPTNeo achieved 95.5% accuracy.
[89]	✓						Random Forest achieved 95–99% across languages.
[68]			✓			✓	RoBERTa achieved F1 = 96%.
[95]	✓	✓					Autoencoder + linear kernel: F1 = 95% (EN), 72% (ES).
[52]			✓		✓	✓	Longformer obtained 68–96% recall.
[54]		✓	✓		✓	✓	CheckGPT outperformed, >99% accuracy.
[90]			✓		✓	✓	MDeBERTa and RoBERTa achieved 93% accuracy.

87% and 88%. However, the fine-tuned LLMs outperformed all other methods, with MDeBERTa and RoBERTa achieving the best results, both reaching an accuracy of 93%.

Table 3 presents an overview of the techniques used by most of the studies, like classical machine learning, deep learning, transformers, existing commercial AI detectors, and statistical approaches. The outperforming methods are presented with a blue tick, and the best results achieved with those techniques are also presented. The table shows that transformers are the most frequently used technique for the task.

A. SHARED TASK

1) ALTA

organized by the Australasian Language Technology Association (ALTA) to detect MGT in the English language [16]. They presented a balanced corpus of 20k, which was described in detail in the corpus section. The winning team [109] in this competition achieved an accuracy of 99%. They implemented the Falcon-7B[34] model that is trained on 15,000B tokens along with label smoothing and a prompting technique.

2) AUTEXTIFICATION

explored the problem not only of detecting MGT vs. HWT but also of identifying the specific model responsible for generating the text [15]. The task consists of two subtasks: (1) binary classification and (2) model attribution. The task is explored in English and Spanish. For subtask 1, the team TALN-UPF reported the best Macro-F1 score of 80% and

[34]https://huggingface.co/tiiuae/falcon-7b Last visited: 01-20-2025

70% for English and Spanish, respectively. Similarly, team Drocks achieved the top position for both languages. The best results obtained were 62% Macro-F1 for English and 65% Macro-F1 for Spanish.

3) CLIN33

shared task presented the corpus for English and Dutch [17]. The team Van Halteren reported the best accuracy for the English corpus set, while team Elsevier stood on top for Dutch by obtaining 85% and 75% accuracy scores, respectively.

4) DAGPAP22

developed corpora for the task of detecting machine-generated scientific articles [20]. The corpus consists of 30,000 human-written scientific articles from ScienceDirect[35] and applied three alteration techniques to the sourced articles. The chunk selected for the alteration consists of characters ranging from 2,500 to 4,000. After implementing these alterations, techniques such as synonym replacement were used using NLTK [110]. Text summarization and paraphrasing were also applied using T5 and ChatGPT, respectively. Reference [111] obtained 99% F1-score by applying an ensemble learning approach. The models that were included in the ensemble are DeBERTaV2 and ALBERTV2.

5) SEMEVAL-TASK8

the 18th International Workshop on Semantic Evaluation announced Task 8 in the category of LLM capabilities [19]. SemEval presented the M4 corpus [85]. The task was further divided into three subtasks. The first task involved detecting MGT vs. HWT for monolingual English text and multilingual text. The second task focused on model attribution, which focused not only on detecting MGT but also on identifying the model that generated that specific chunk. The third task involved boundary detection of MGT vs. HWT. Reference [85] presented the M4 corpus and developed multiple detectors such as RoBERTa, ELECTRA, XLM-R, logistic regression with GLTR features, and SVM with NELA features on the proposed corpus. The RoBERTa model achieved a perfect accuracy of 100%, demonstrating its robustness across languages and domains. Metrics such as cross-domain precision and recall were used, and the findings highlighted the importance of multilingual adaptability in developing universal text detection systems. Moreover, human evaluation was conducted on a subset of Reddit and arXiv by asking six skilled annotators. The average score achieved by humans was 77% for Reddit and 72% for arXiv.

The first position was achieved for the first task of the monolingual track by team Genaios [112]. They focused on developing a machine-generated text detection system for both monolingual and multilingual corpora. The authors used LR with TF–IDF and readability[36] features, RoBERTa and e5, and also fine-tuned GPT-2 and LLaMA-2. The evaluation metrics included accuracy, precision, and recall. For monolingual tasks, the system achieved an accuracy of 96.88%, while for multilingual tasks, the accuracy was 89.97%.

For the multilingual track, team USTC-BUPT [113] achieved the highest accuracy. This team addressed all three subtasks by developing techniques named DATeD, LLAM, TLE, and AuDM. These techniques were developed as DATeD for monolingual Task A, LLAM for multilingual Task A, TLE for Task B, and AuDM for Task C. DATeD incorporated a gradient reversal layer along with the RoBERTa model, while LLAM first extracted an embedding that later passed on to the CNN model for classification. TLE utilizes the LLM Embeddings method with a minor modification in the classification strategy. In contrast, AuDM employs data augmentation and fine-tunes the DeBERTa model. The best accuracy scores achieved by Task A for monolingual and multilingual were 96.10% and 95.97%, respectively. Similarly, the study ranked third for Tasks B and C by achieving an accuracy of 84.33% for Task B and an MAE of 17.702 for Task C. The study concluded that domain-adaptive models can bridge gaps between corpora with differing linguistic characteristics, making them suitable for real-world applications.

For subtask B, the team AISPACE [114] ranked at the top. They evaluated pre-trained transformer models and proposed three techniques: encoder only, decoder only, and encoder–decoder. The encoder models were RoBERTa, DeBERTa, and Longformer. XLNet is used for the decoder, while T5 is used as the encoder–decoder. The system achieved the best results with RoBERTa, achieving an accuracy of 99.46% for multi-way classification of text by detecting the MGT and attributing the model that generated that text.

The top-ranked team for subtask C, which is boundary identification, the TM-TREK team [115] emerged as the top performer in the SemEval task. The problem was explored using transformers like XLNet, BigBird, and Longformer for detecting MGT mixed with HWT. The transformers were further enhanced with layers of deep learning models, such as LSTM, to generate better and more coherent text. The evaluation was conducted using Mean Absolute Error (MAE), with the best result being an MAE of 2.44, achieved on XLNet, indicating high accuracy in domain-specific tasks.

Gen-AI: is the most recent shared task for the problem of detecting MGT [116]. The task consisted of three tracks to handle the problem diversely. Task 1 simply identified the MGT for English and multilingual corpora. Task 2 handled the academic essay authenticity challenge for English and Arabic. Finally, Task 3 explored the robustness of the system for cross-domain detection of MGT for both adversarial and non-adversarial settings.

For Task 1 [21], first place was taken by team Advacheck [117], achieving an F1-score of 83% on

[35]https://www.sciencedirect.com/ Last visited: 01-18-2025
[36]https://github.com/andreasvc/readability/ Last visited: 01-20-2025

TABLE 4. Summary of Methods for Shared Tasks. CML, DL, Trans, and EL refer to classical machine learning, deep learning, transformer-based models, and ensemble learning, respectively.

Reference	CML	DL	Trans	EL	Best Results
[110]			✓		Falcon-7B achieved 99% accuracy.
[17]	✓		✓	✓	Obtained 85% and 75% accuracy for English and Dutch.
[15]		✓	✓		BiLSTM and LLMs achieved 80% / 70% F1 for English / Spanish.
[112]			✓	✓	Ensemble approach achieved an F1 of 99%.
[86]	✓		✓		RoBERTa obtained 100% accuracy.
[113]	✓		✓		LLaMA-2 obtained 96% accuracy.
[114]		✓	✓		CNN achieved 95% accuracy.
[115]			✓		RoBERTa achieved 99% accuracy.
[116]		✓	✓		XLNet mean absolute error = 2.44.
[118]	✓		✓		DeBERTa-v3-base achieved 83% accuracy.
[119]			✓	✓	Ensemble approach achieved 79% F1.
[122]			✓		Reported 99% F1 utilizing LLaMA.
[121]			✓		Fine-tuned AraELECTRA outperformed by achiving 98% F1.
[123]			✓		Distil-RoBERTa achieved 99% F1.

the encoder model, i.e., DeBERTa-v3-base, for English. In multilingual settings, team Grape [118] won first place and achieved a 79% F1-score. The model employed by the team was an ensemble model from the families of LLaMA and Gemma.

Task 2 [119] for the essay authenticity challenge, team CMI-AIGCX [120] and team IntegrityAI [121] ranked on top for English and Arabic, respectively. CMI-AIGCX reported a 99% F1-score utilizing LLaMA, while IntegrityAI achieved a 98% F1-score by fine-tuning AraELECTRA.

Team Leidos [122] ranked on top for both subtasks of Task 3 [123]. The team trained four different models of Distil-RoBERTa to achieve the best results. The best results reported for the subtasks of cross-domain detection and adversarial robustness are 99% and 97%, respectively.

To summarize, classical machine learning, deep learning, and transformer-based models are used to achieve better results on different corpora for the task of detecting MGT. Moreover, Table 4 presents the trend for shared tasks.

VI. EXPERIMENTAL SETUP

To develop a reliable model, it is important to select a well-defined experimental setup [124]. The experimental setup determines how the Corpus is divided into training and testing sets, ensuring a fair evaluation of the model's performance. The model should not be evaluated on the data used in the training process to be deployed in the real world, as this could lead to unrealistic results and poor generalization in real-world scenarios. The two main data partitioning strategies used by researchers for the task of detecting MGT are:

- ***Train–Test Split:*** In the train–test split, the corpus is split into two sets, one for the learning of the model and the other to evaluate the trained model. Most studies on the detection of HWT vs. MGT have employed this evaluation methodology.
- ***k-Fold Cross-Validation:*** Cross-validation divides the corpus into k sets, which is why it is called k-fold cross-validation. In this evaluation, the training process works in a loop. Each set of the corpus can be used for training and testing, but not at the same time. This approach facilitates having multiple splits. Only a few studies have implemented the cross-validation approach, as it requires more time and resources [82], [94].

VII. EVALUATION MEASURE

Evaluating a model is the most critical step in assessing its effectiveness. The choice of evaluation metrics directly impacts the interpretation of the model's performance. In the task of MGT detection, both human and automated evaluation methods are utilized to assess the reliability and effectiveness of the models. Table 5 presents the distribution of studies that have adopted these evaluation measures for MGT detection systems. This section explains why these evaluation metrics are used and their significance in this domain.

A. HUMAN EVALUATION

Human evaluation is the process in which domain experts are employed to check whether the text is human-written or machine-generated. It provides qualitative insights that automated metrics may overlook, such as coherence, fluency, and logical consistency. Different studies have employed different human evaluation techniques to measure human performance in distinguishing between HWT and MGT [24], [57], [85]. Gambetti et al. [57] conducted a human evaluation with 80 participants, who were asked to select whether the text is machine-generated or indicate if they were unsure about its authorship. The "not sure" option was added to reduce random guessing. The human evaluation resulted in an accuracy of 57.13%, leading to the conclusion that humans are not highly capable of distinguishing HWT from MGT. Similarly, Wamg et al. [85] asked six proficient annotators to classify the sample of 50 instances as HWT and MGT, achieving a best accuracy of 77%, which was still lower than that of an automated detector, RoBERTa, that achieved 100% accuracy. Finally, Ali et al. [24] examined human evaluation at different textual levels,document, paragraph, and sentence,finding that human performance improved on smaller text segments. These findings highlight the limitations of human evaluation, especially for large-scale detection tasks, reinforcing the need for automated evaluation metrics.

In short, humans and models catch different things. Trained detectors are great for fast, large-scale screening when the data looks familiar. Humans do better when the text is unusual, context-heavy, or needs fact checks. The sweet spot is a hybrid: let a model triage, escalate low-confidence or high-stakes cases to a human, and keep thresholds tuned as data drifts.

B. AUTOMATED EVALUATION

Automated evaluation metrics provide objective and scalable means of assessing model performance. Given that distinguishing between HWT and MGT is a binary classification problem, researchers commonly employ several standard classification metrics such as accuracy, precision, etc. These metrics are chosen based on their ability to reflect different aspects of model effectiveness:

- **Accuracy:** The ratio of correctly predicted instances out of all cases. It reflects the overall effectiveness of the model, but may be misleading in imbalanced corpora where one class dominates. For instance, the Myle Ott corpus [31] contains 800 human-written while 3,200 machine-generated instances, which reflects that the data are highly imbalanced.
- **Precision:** The proportion of true positive predictions (TP) among all positive predictions ($TP + FP$). High precision is crucial in applications where false positives (incorrectly classifying human-written text as machine-generated) must be minimized, such as in academic integrity checks.

TABLE 5. Number of studies using each evaluation measure.

Evaluation Measure	No. of studies used
Human Evaluation	3
Accuracy	24
Precision	11
Recall	12
F1-Measure	28
Area Under Curve (AUC)	8
Mean Absolute Error (MAE)	2

- **Recall:** The proportion of true positive predictions (TP) among all actual positives ($TP + FN$). High recall is essential in scenarios where missing MGT instances (false negatives) are costly, such as in misinformation detection.
- **F1-score:** The F1-score is the harmonic mean of precision and recall $\left(2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}\right)$. It is a useful metric for classification tasks, particularly in the case of imbalanced corpora, as it provides a balance between precision and recall.
- **Area Under Curve (AUC):** Represents the area under the Receiver Operating Characteristic (ROC) curve. A higher AUC indicates that the model consistently ranks positive instances above negative ones.
- **Mean Absolute Error (MAE):** Measures the average absolute difference between predicted and actual values, making it useful for assessing probabilistic outputs and model calibration.

VIII. PRACTICAL IMPLICATIONS AND REAL-WORLD USE CASES

The ability to accurately detect machine-generated text (MGT) has moved from a theoretical challenge to a practical necessity with significant real-world implications. This section explores key use cases and the ethical and policy considerations that come with implementing detection systems.

A. APPLICATIONS IN KEY DOMAINS

The applications of MGT detection are expanding rapidly across various sectors, driven by the need for authenticity, transparency, and integrity.

B. EDUCATION AND ACADEMIC INTEGRITY

In education, the proliferation of LLMs poses a significant challenge to academic integrity. MGT detection is a crucial tool for identifying potential instances of plagiarism and academic dishonesty. Detection systems help instructors and academic institutions ensure that submitted work reflects a student's own effort and understanding rather than a machine's. This is particularly important for high-stakes assessments like essays and research papers. However, detection in this domain must be used with caution, as false positives could unfairly penalize students, especially those

from diverse linguistic or neurodivergent backgrounds whose writing styles may be misclassified.

C. JOURNALISM AND MEDIA

The media landscape is increasingly vulnerable to the spread of misinformation and disinformation, which can be generated at scale by LLMs. Journalists and news organizations can use MGT detection to verify the authenticity of sources, identify synthetic news articles, and protect their platforms from malicious actors. Detection tools act as a first line of defense against ''fake news,'' helping media outlets maintain trust with their audience.

D. CONTENT MODERATION AND SOCIAL MEDIA

For social media platforms, MGT detection is a vital component of content moderation efforts. It helps to identify and remove malicious content like hate speech, spam, propaganda, and misinformation campaigns, which can now be produced in enormous volumes by AI. By flagging suspicious content, MGT detection systems assist human moderators, making the process more efficient and helping to curb the spread of harmful narratives.

1) ETHICAL, LEGAL, AND POLICY IMPLICATIONS

As MGT detection becomes more commonplace, it introduces complex ethical, legal, and policy challenges that must be carefully managed.

a: THE RISK OF BIAS AND FALSE POSITIVES

A major concern is algorithmic bias. If detection models are trained primarily on content from a limited demographic or cultural group, they may unfairly flag texts from other groups as ''unnatural'' or machine-generated. This could disproportionately affect non-native speakers, authors with unique writing styles, or even works from different time periods. False positives pose a significant ethical risk, as they could lead to unjust accusations of academic misconduct or content removal.

b: ACCOUNTABILITY AND LEGAL FRAMEWORKS

The rise of MGT and detection systems complicates issues of accountability. Who is responsible when an MGT detector fails? Is it the developer of the detector, the user who relied on it, or the creator of the MGT? As of now, there is a significant lack of legal and regulatory frameworks for MGT and its detection. This calls for the development of clear policies to govern the responsible use of AI and to establish accountability for its outputs.

2) THE ROLE OF EXPLAINABLE AI (XAI)

Given the high-stakes nature of MGT detection, particularly in academic and legal contexts, transparency is paramount. This is where Explainable AI (XAI) comes in. XAI systems aim to make the decisions of a ''black-box'' model understandable to humans. For MGT detection, an XAI system could not only classify a text as machine-generated but also provide a reason for the classification. For example, it could highlight specific phrases with low perplexity, repeated syntactic structures, or a lack of emotional markers that led to its conclusion. This provides a human user with a basis for trust and helps them make a final, informed decision rather than blindly following an algorithm.

IX. DISCUSSION

The survey of human-written text (HWT) versus machine-generated text (MGT) detection reveals several recurring limitations in both the underlying corpora and the detection methodologies. Addressing these gaps is critical to advancing more robust, generalizable, and equitable detection systems. First, existing corpora exhibit narrow topical coverage: most corpora focus on reviews, question–answer pairs, or social media posts, while creative writing, medical narratives, and other domains remain underrepresented. This confinement restricts a model's ability to generalize beyond its training distribution. Furthermore, language diversity is insufficient. Major world languages such as Hindi, Bengali, Portuguese, and Urdu (each among the top ten most spoken globally[37]) are rarely included in MGT corpora, hindering applicability in multilingual and low-resource settings. Many corpora rely heavily on prompt engineering with a small subset of LLMs, predominantly GPT-family models, to generate and paraphrase text. As shown in Figure 2, most of the corpora were developed using a limited number of LLMs and mainly used models from the GPT family, so there is a need to use open-source models like CODEX, Alpaca, Koala, FLAN, Zephyr, DeepSeek, MT5 [125], etc. Moreover, many corpora rely on only a few language models, so they don't fully capture the variety of styles and nuances in MGT. Additionally, corpora often suffer from class imbalance, where certain languages or genres dominate the sample, biasing the trained detectors. Finally, nearly all surveys treat detection as a binary classification problem, with few addressing realistic adversarial scenarios like paraphrasing or mixed human–machine edits, which are increasingly common in practice.

In terms of techniques, the field has trended toward transformer-based detectors, which achieve strong performance but require substantial computational resources, limiting their accessibility. Classical machine learning and lighter deep learning architectures have demonstrated competitive accuracy in some settings, yet they remain overshadowed by large transformer-based architectures, despite their efficiency advantages. Moreover, the focus on binary detection hampers progress on multi-class adversarial robustness, where systems must identify not only MGT but also the nature of machine involvement, whether text is polished, paraphrased, or deeply mixed with human writing.

In practice, there's a straightforward trade-off. Transformer detectors (e.g., RoBERTa, Longformer) are usually the most

[37] https://www.ethnologue.com/insights/ethnologue200/ Last visited: 01-19-2025

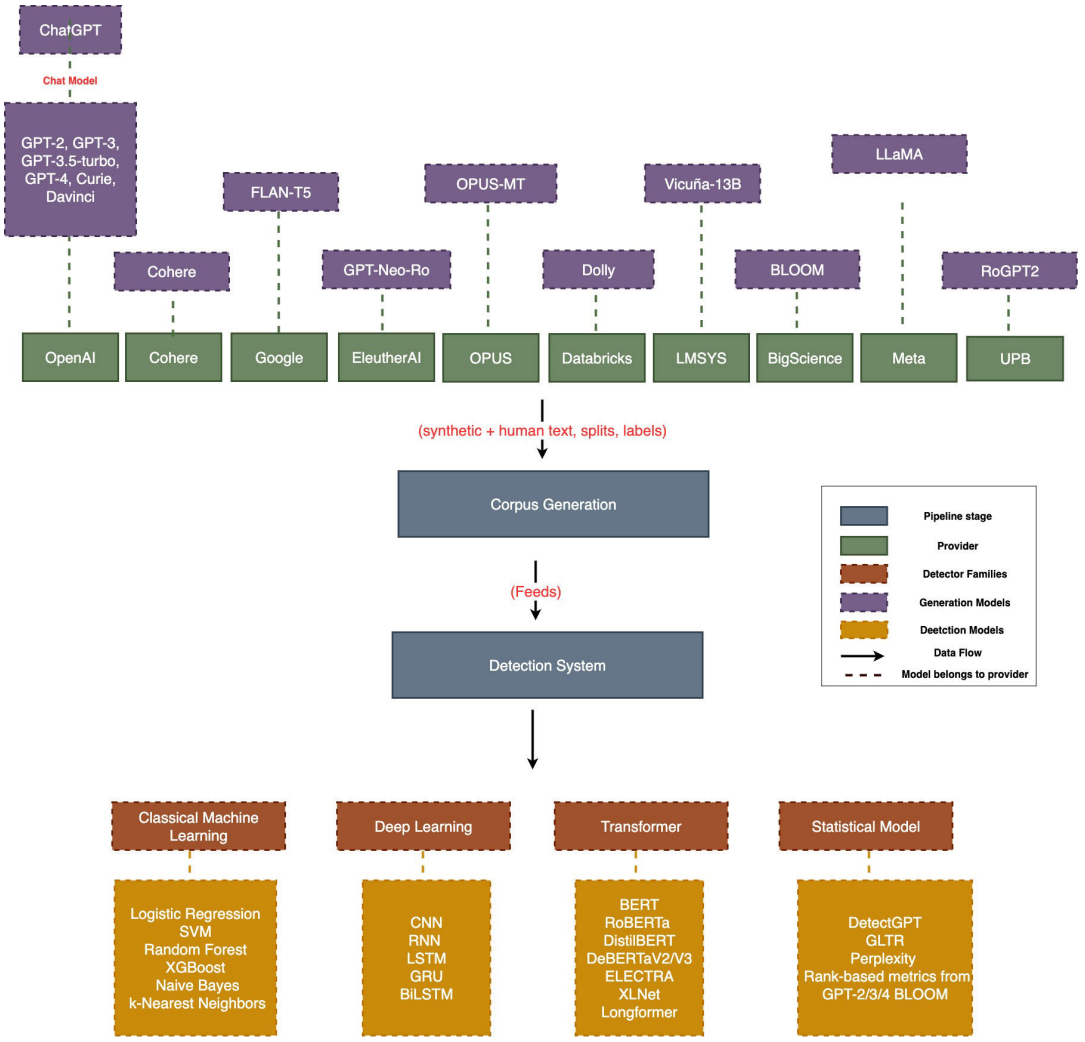


FIGURE 2. Model ecosystem: commonly used models for corpus generation and downstream MGT detection.

accurate and handle long, multilingual text, but they're costly to run, slower, and harder to explain or calibrate. Classic feature-based methods (TF–IDF, stylometry, perplexity/rank) are fast, cheap, and transparent, yet they struggle on long, mixed, or open-domain inputs and often need language-specific tweaks. A practical middle ground is a two-stage setup: use a lightweight classic screen for most cases and send only uncertain ones to a transformer,backed by adversarial training data, regular threshold tuning to a target FPR, and model distillation/quantization to meet latency and privacy constraints.

These limitations suggest several directions for future research. Expanding corpus diversity, both in domain and language, is paramount; researchers should curate or adapt corpora for underrepresented genres and low-resource languages. Broadening the suite of source LLMs and balanced sampling strategies will mitigate model and language biases. Methodologically, pursuing lightweight detection architectures and modular pipelines can democratize MGT detection in constrained environments. Finally, developing adversarial and multi-class benchmarks will push detectors beyond binary classification and more closely align evaluation with real-world challenges. By confronting these gaps, the community can build on the foundations surveyed here to develop detection systems that are not only accurate but also adaptable, resource-efficient, and fair across languages and domains.

X. LIMITATIONS

While MGT detection has come a long way, several limitations remain,and they point to clear directions for future work.

A. SELECTION BIAS AND LIMITED GENERALIZATION

Many studies train and evaluate on a narrow slice of public data (e.g., Wikipedia, news) and a small set of generators (often a single model like GPT-3). This narrow focus limits how well detectors generalize. A system that performs well on one benchmark may stumble on a new domain (creative

writing, legal text), a different topic, or outputs from a different LLM.

B. ADVERSARIAL ROBUSTNESS AND EVASION

This is an ongoing arms race. Simple tactics,paraphrasing, light human edits, or switching to less common models,can still slip past many detectors. Building systems that stay reliable under these evasions and adapt to unseen MGT styles remains a core challenge.

C. MULTILINGUALITY AND LOW-RESOURCE LANGUAGES

Most research focuses on English. High-quality corpora are scarce for many other languages, especially low-resource ones. That gap matters because the cues used for detection,like morphology and syntax,are language-specific. Expanding coverage beyond English is essential.

XI. CONCLUSION AND FUTURE DIRECTIONS

In this survey, we have presented a comprehensive overview of the field of machine-generated text detection, providing a critical analysis of its foundational linguistic frameworks, the corpora used for research, and the methods developed for detection. Our work moves beyond a simple review, offering a new theoretical taxonomy of linguistic features and a critical evaluation of the field's key challenges.

The core contribution of this paper is a new unifying framework that distinguishes between formal and pragmatic linguistic features. Our analysis demonstrates that while the field has made significant progress in detecting formal features, a critical gap remains in the detection of nuanced human expression. This gap is a direct result of the continuous advancement of generative models, which are making their outputs increasingly difficult to distinguish from human-written text.

Based on our findings, we propose a clear agenda for future research.

1) The field must move beyond English-centric corpora and develop robust datasets for low-resource languages to ensure that detection models are fair and effective globally.
2) Future models must be designed to withstand evasion techniques such as paraphrasing and human-in-the-loop editing. This may require moving beyond single-shot detection to more continuous, behavioral monitoring of text.
3) For detection systems to be widely adopted and trusted in high-stakes environments, they must be transparent. Future research should focus on integrating XAI to provide human-readable explanations for a model's classification, which will build user trust and reduce the risk of false positives.

Ultimately, the future of machine-generated text detection lies in creating systems that are not only accurate but also robust, fair, and transparent. We hope this survey provides a valuable roadmap for researchers and practitioners working to address one of the most pressing challenges in the digital age.

ACKNOWLEDGMENT

The authors are grateful to the reviewers for their time and invaluable, constructive feedback, which helped improve the quality of the article.

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